Breast Tumor Classification with Random Forest &

Diagnostic Pattern Segmentation

UCI Wisconsin Diagnostic Breast Cancer (WDBC) Dataset

TITANS · Healthcare Analytics Capstone · 2026

Team: Anjaneyulu Pathuri · Susmit Sudhakaran · Saikiran Malleli

Rittika Bhattacharya · Vaibhav Patil

Model: Random Forest Classifier · Dataset: WDBC (569 records)

"Focused on risk prioritization, interpretability, and decision-support analytics."

OUR TEAM

Meet Team Titans

Roles were collaboratively shared across all team members

AP

Anjaneyulu Pathuri

From: Andhra Pradesh
Domain: Software Development
Profession: Tech Lead
14+ Yrs Exp

SS

Susmit Sudhakaran

From: Maharashtra
Domain: Fintech
Profession: Sr. Data Engineer
9+ Yrs Exp

SM

Saikiran Malleli

From: Andhra Pradesh
Domain: ERP, SUPPLY CHAIN
Profession: Product Manager
8 Yrs Exp

RB

Rittika Bhattacharya

From: West Bengal
Domain: Teaching & Education
Profession: Asst Professor of Statistics
5 Yrs Exp

VP

Vaibhav Patil

From: Maharashtra
Domain: Oil & Gas
Profession: Sr. Data Quality Analyst
12+ Yrs Exp

BUSINESS CONTEXT

Business Problem & Objective

THE CHALLENGE

1. **Large-volume diagnostic data**

   Healthcare screening generates vast, hard-to-interpret FNA measurement sets across 30 features.

2. **Inconsistent risk flagging**

   Manual review leads to variability in identifying high-risk malignant cases.

3. **Missed malignant cases**

   False negatives are the most critical error — a missed malignant = missed investigation.

PROJECT OBJECTIVE

- ✓ Classify tumors as Malignant or Benign using Random Forest
- ✓ Maximize sensitivity - protect malignant recall above all
- ✓ Perform clustering analysis to identify hidden data patterns
- ✓ Build a reproducible, governed ML workflow

PRIMARY SUCCESS METRICS (Charter-Defined)

ROC-AUC Primary model ranking metric

Recall (Malignant) Operational safeguard — minimize FN

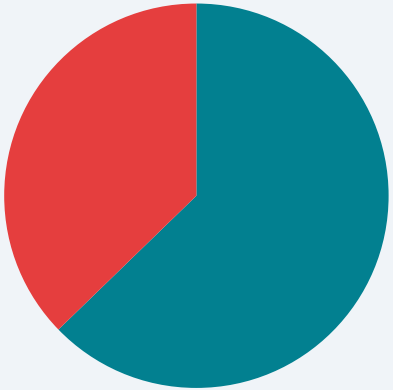
F1-Score (Malignant) Threshold-balancing metric

⚠ *Educational & analytical in nature — not intended for clinical diagnosis or medical advice.*

DATA UNDERSTANDING

Dataset Overview & Key Observations

569	30	357	212
Total Records	Diagnostic Features	Benign Cases (B)	Malignant Cases (M)



Benign — 357 (62.7%) Malignant — 212 (37.3%)

FEATURE GROUPS · FNA-derived measurements

Radius / Texture / Perimeter
Core shape dimensions from cell nuclei images

Area / Smoothness / Concavity
Size, surface uniformity & irregularity indicators

Compactness / Symmetry
Shape boundary characteristics

Fractal Dimension
Structural regularity of cell boundaries

Mean · SE · Worst (×10 each = 30 total)
Three statistical variants per feature family

✓ *No missing values · No duplicates · Clean schema validated · Outliers retained (RF robust to extremes)*

Data Validation & Preparation Strategy

VALIDATION ACTIVITIES

- ✓ Renamed columns to documented WDBC feature names
- ✓ Removed non-informative ID column
- ✓ Confirmed 30 features after ID & label exclusion
- ✓ Validated zero missing values across all records
- ✓ No duplicate rows found in 569 records
- ✓ Target encoded: Malignant (M) = 1, Benign (B) = 0

DATASET SPLIT (Stratified)

Train (70%) — 397 records
Model learning · stratified to preserve class ratio

Validation (15%) — 86 records
Hyperparameter tuning & threshold selection only

Test (15%) — 86 records
Final one-time evaluation · fully isolated throughout

EVALUATION INTEGRITY — LEAKAGE PREVENTION

→ All preprocessing, tuning & threshold decisions made exclusively on train + validation data

→ Test set reserved for final one-time reporting — split indices saved for full reproducibility

METHODOLOGY

Risk Identification Workflow

WHY RANDOM FOREST?

- ✓ Handles nonlinear relationships in tabular data
- ✓ Tolerates correlated predictors effectively
- ✓ Ensemble averaging reduces overfitting risk
- ✓ Provides feature importance for interpretation
- ✓ Charter-mandated as the sole supervised algorithm

01

Diagnostic Data Input (WDBC · 569 records)

02

Data Preparation & Validation

03

Random Forest Classifier

04

Hyperparameter Tuning (GridSearchCV)

05

Threshold Optimization & Final Evaluation

Stability **Reproducibility** **Risk Sensitivity** **Interpretability**

OPTIMIZATION

Model Optimization & Tuning

Base Model (200 estimators, max_depth=None)

GridSearchCV (5-fold stratified, ROC-AUC refit)

Improved Stability & Malignant Recall

Tuned Config — max_depth=10, FN: 3→2

FINAL TUNED CONFIGURATION (GridSearchCV)

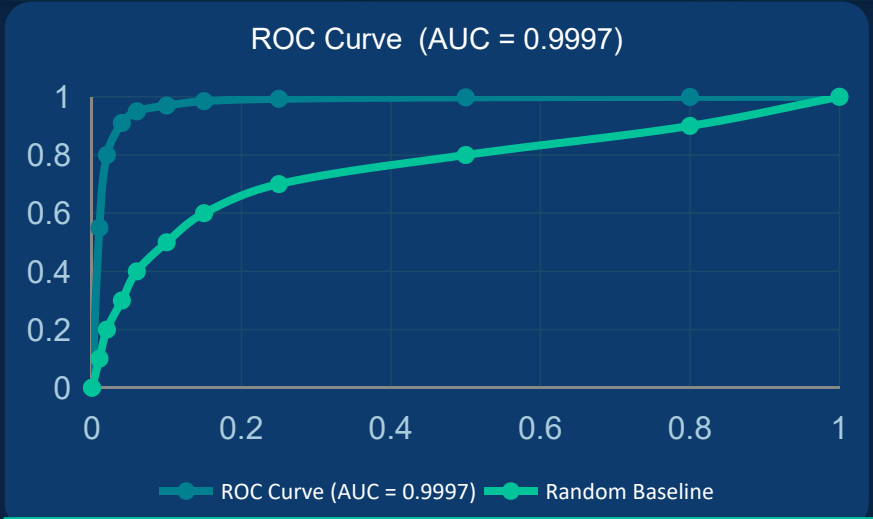
Parameter	Value	Purpose
n_estimators	100	Tree count for stability
max_depth	10	Prevent overfitting (was None)
min_samples_split	2	Split threshold
min_samples_leaf	1	Leaf size control
max_features	sqrt	Feature subset per split
class_weight	None	Imbalance handled via threshold

Tuning Result: Validation FN reduced from 3 → 2 · Malignant Recall improved · Overfitting controlled

RESULTS

Final Performance Results

96.51%	1.00	0.91	0.95	0.9997
Test Accuracy	Precision (Malignant)	Recall (Malignant)	F1-Score (Malignant)	ROC-AUC



CONFUSION MATRIX (Test Set — 86 records)

	Pred: Benign	Pred: Malignant
	TN 54 True Negative	FP 0 False Positive
	FN 3 False Negative	TP 29 True Positive

All 54 benign cases correctly identified · 29/32 malignant correctly detected · 3 FN motivated threshold optimization

Threshold Calibration & Error Analysis

OPTIMAL THRESHOLD

0.2800

Default threshold: **0.50**
After optimization: **0.28**

Result: 0 False Negatives
Accuracy: 98.84%

OPERATIONAL TRADEOFF FRAMEWORK

FALSE NEGATIVE (Highest Risk)

Before: 3 FN on test set
After threshold: 0 FN
→ Priority: Eliminate these

FALSE POSITIVE (Acceptable)

0 FP on initial test set
Borderline benign may flag
→ Adds review effort only

ROOT CAUSE — ERROR PATTERN ANALYSIS

Missed Cases (FN)	Feature values in overlapping benign/malignant boundary regions
Threshold Effect (0.28)	Eliminated all 3 FN on test set — accuracy rose from 96.51% to 98.84%
Hard Cases (FP risk)	Benign tumors with morphology near the shifted decision boundary
Top Features (Error)	radius_worst, perimeter_worst, concave_points_worst drove most decisions

SEGMENTATION

Diagnostic Pattern Segmentation

Classification

"What is this case?"

+

Segmentation

"What behavior groups exist?"

= **Deeper Interpretability**

METHODS & PREPARATION

Preprocessing
- Diagnosis label excluded
- All 30 features StandardScaled
- 569 rows × 30 features matrix

K-Means Clustering
Optimal k via Elbow Method

Hierarchical Clustering
Validated via Dendrogram

PCA Visualization
PC1: 44.27% · PC2: 18.97%
Combined: 63.24% variance

LOW RISK

Cluster 1

Lower-Risk Pattern Group

Predominantly benign morphology
Smaller radii, lower concavity
Mostly benign labels align

MIXED

Cluster 2

Borderline / Mixed Group

Feature values overlap between benign & malignant patterns
Requires closer analytical review

HIGH RISK

Cluster 3

Higher-Risk Pattern Group

Malignant-leaning morphology
Higher concavity & texture scores
Most malignant labels found here

Exploratory pattern segmentation only — not medical subtyping or clinical treatment guidance

Key Insights & Strategic Recommendations

Insight 01

Threshold > Raw Accuracy

Lowering threshold to 0.28 eliminated all 3 FN on test set — accuracy jumped from 96.51% to 98.84%.

Insight 02

ROC-AUC Near Perfect

0.9997 ROC-AUC confirms excellent probability ranking — the model can reliably separate classes.

Insight 03

Segmentation Adds Depth

PCA + K-Means revealed interpretable morphology clusters beyond binary classification output.

STRATEGIC RECOMMENDATIONS

Use Threshold-Tuned Model
Present the 0.28-threshold model as the preferred final output — 98.84% accuracy, 0 FN.

Strict Test-Set Governance
Maintain one-time test usage discipline; all future threshold/parameter decisions on train+val only.

Present Clusters Descriptively
Use cluster profiles as morphology communication tools, not as medical subtypes.

Expand Dataset Diversity
One benchmark source limits generalizability — multi-institutional data needed for real deployment.

Add Explainability Layer
Incorporate SHAP/LIME to surface feature-level reasoning for each prediction made.

FUTURE WORK

Limitations & Future Enhancements

CURRENT LIMITATIONS

Small Dataset (569 records)

Limits statistical generalizability to broader hospital populations.

Single Benchmark Source

No external hospital validation — not representative of real deployment.

No Clinical Validation

Analytical workflow only — not tested in a clinical setting.

No Imaging / Genetic Data

Restricted to FNA measurements; no additional patient variables.

Overfitting Risk (Initial)

Base model with max_depth=None showed perfect training performance.

FUTURE ENHANCEMENTS

1. **External Validation**

   Test on independent hospital datasets to verify generalizability.

2. **Explainable AI (SHAP/LIME)**

   Surface feature-level reasoning for every individual prediction.

3. **Human-in-the-Loop**

   Add clinical reviewer gates for borderline probability cases.

4. **Real-time Drift Monitoring**

   Detect data and model degradation in production pipelines.

5. **Probability Calibration**

   Platt scaling / isotonic regression for better-calibrated outputs.

⚠ *This workflow is for learning & demonstration only — not intended for clinical diagnosis or medical deployment.*

Thank You

Questions & Discussion

TEAM TITANS

- Anjaneyulu Pathuri
- Susmit Sudhakaran
- Saikiran Malleli
- Rittika Bhattacharya
- Vaibhav Patil

PROJECT AT A GLANCE

ROC-AUC: **0.9997**

Final Accuracy: **98.84%**

FN after threshold: **0**

Threshold: **0.28**

Records: **569 WDBC**